

KITEWRIGHT RANGE · REQUIREMENTS

Kitewright Range requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status from KWR-CAL-001 v0.2 on the constructable design; R10 cost stated against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 concept status for every requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Range requirements

Six of the ten requirements are met on paper or by design. Endurance (R2) and take-off mass (R7) are not met, and hover thrust margin (R1) and survey range (R3) are met by margins too thin to rely on until the motors and propellers are measured (KWR-CAL-001 v0.2, Table 6). The targets below are unchanged from the scaffold; what to do about each shortfall is an open decision for Amish (docs/06-design-decisions.md).

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (KWR-CAL-001)
R1	Vertical take-off and landing at high altitude	Hover with at least 30% thrust margin at 0.74 kg/m ³ air density (5,000 m) at maximum take-off mass	Lift motor and propeller thrust measured in AltiRig at 5,000 m density; margin calculated for the as-built mass	At risk: 31.8 % on maker-class thrust at 10.58 kg
R2	Cruise endurance at altitude	At least 45 min wing-borne flight at 5,000 m and -20 C with a 1 kg payload and 20% reserve (target)	Energy model checked against bench power data, then timed flight at a high site	Not met: 31.0 min
R3	Survey range	At least 40 km of survey track per flight at 5,000 m (target)	Flight log distance on a mapped survey pattern	At risk: 40.9 km
R4	Payload capacity	1.0 kg on the Kitewright mount within the centre of gravity range	Weigh and balance check; flight with a dummy payload	Met by design: mount centred on the CG, 28.4 % of the mean chord with or without the payload
R5	Operating temperature	-20 to +40 C (-4 to +104 F) for all airframe electronics and servos	Cold chamber soak and function test of servos, motors and avionics	Met on paper by specification; cold ratings to confirm when parts are bought
R6	Wind tolerance	Take off, transition and land in sustained wind of 10 m/s with gusts to 14 m/s	Flight tests with a logged anemometer at the launch site	Met on paper: cruise 21.9 m/s, stall 17.6 m/s, about 9 degrees of tilt holds a 14 m/s side gust in hover
R7	Maximum take-off mass	No more than 8 kg including payload (target)	Weigh the complete aircraft	Not met: 10.58 kg
R8	Transport	Packs into cases of no more than 1.3 m long; each carried load no more than 15 kg	Pack and measure; two-person carry trial	Met by design: longest part 1,205 mm; two cases of about 9 and 12 kg
R9	Patent design-arounds held	Fixed motors, single wing, rotor discs clear of the wing-top plane in every revision	Design review checklist at each revision	Met by design: discs 81 mm below the wing-top plane, checked by the model
R10	Open and buildable	All parts off the shelf or made with hobby tools; bill of materials at or below USD 5,000	Costed BOM review; independent build by a second team	Met on paper: USD 3,945 against the USD 5,000 value-engineering target (USD 1,055 under)

■ Requirements at risk or not met

- R2 and R7 pull against each other: more pack energy adds mass, and a lighter aircraft carries less energy. The options are worked out in KWR-CAL-001 section H.
- R1 rests on maker-class thrust that has not been measured at 5,000 m density.
- R3 follows R2; any change that lengthens endurance lengthens range.

■ Assumptions

- Hover and cruise figures are at the ISA density for 5,000 m (0.736 kg/m³) with the ColdCell packs warmed before take-off.
- Stock PX4 quadplane firmware is enough; no custom flight control code is needed (KWR-DDR-001, D1).
- ColdCell packs deliver 95 % of their capacity at -20 °C once warmed (cross-repo assumption).
- Most survey missions can start from a site within 10 km of the target lake.